



**Politecnico
di Torino**

Nuclear Fission Plant
Sustainable Nuclear Energy

Assignment 2

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1 General introduction

In this project it is studied the AP1000 reactor designed by Westinghouse, based on AP600 design. As a Generation III+ reactor, the AP1000 incorporates a significant number of passive safety systems aimed at enhancing the overall safety of the plant. The objective of this study is to verify the reliability of the system with the main thermal-hydraulic safety limits. The main goals are the evaluation of the the axial temperature profile of the fuel pellet centreline and the Departure from Nucleate Boiling Ratio (DNBR). These parameters are assessed to ensure that operational conditions remain within acceptable physical boundaries. In particular, the peak value of the fuel centreline temperature profile is compared to the melting temperature of uranium dioxide, while the DNBR is compared to the minimum allowable value recommended by Westinghouse.

2 Assumptions

It is assumed the steady-state condition with only the 97.4% of the nominal power deposited inside the fuel.

The analysis is carried out considering as the hot channel of the core the central sub-channel one, as it is at the beginning of the life (BOL), the region exposed to the highest neutron flux because of its proximity to the core centerline. Since the neutron flux distribution and consequently the power density distribution is:

$$q_v(r, z) = q_{v,max} \cdot J_0 \left(2.4048 \frac{r}{R_e} \right) \cdot \cos \left(\frac{\pi z}{H_e} \right) \quad (1)$$

where R_e and H_e are the extrapolated lengths and J_0 is the order zero Bessel function. Therefore the hot channel volumetric heat generation rate is found imposing $r = 0$, obtaining:

$$q_{v,HC}(z) = q_{v,max} \cdot \cos \left(\frac{\pi z}{H_e} \right) \quad (2)$$

Where $z = 0$ is imposed at half of the core height.

The $q_{v,max}$ is the volumetric heat generation rate at the hot spot, computed starting from the average value and the hot channel factor. This parameter able to represent the entity of non-uniformities with respect to the nominal average value.

In order to complete the assignment it is followed this procedure because if the thermal limits are respected for the hot channel and in particular for the hot spot, the design is adequate throughout the core. Moreover, since computing average values is simpler, in a preliminar design, this is done assuming uniform quantities and applying those correction coefficients.

3 Main procedure

All the following steps were carried out by implementing a code in Matlab and using the XSteam library for the water properties.

To obtain the fuel pellet centerline temperature profile, a backwards procedure is applied starting from the increase of water enthalpy in the hot channel and step by step retrieving the temperature profiles of the water, the cladding and the pellet.

On the other hand, to find the MDNBR, correlations for the critical heat flux are needed and in order to understand the interaction between the cladding and the water, flow quality and void fraction must be computed.

From the "AP1000 Design Control Document" design data are extracted:

Parameter	Symbol	Value
Reactor core heat output	P_{th}	3400 MWt
Heat generated in fuel	-	97.4%
Nominal system pressure	p_{bar}	155 bar
Active fuel height	H_{act}	4.267 m
Number of fuel assemblies	$N_{f,ass}$	157
Fuel rods per assembly	$N_{f,rod}$	264
Total fuel rods	$N_{tot,f,rod}$	41,448
Fuel rod outer diameter	D_{co}	9.5 mm
Clad thickness	-	0.5715 mm
Fuel rod inner diameter	D_{ci}	8.357 mm
Diametral gap	D_{gap}	0.165 mm
Pellet diameter	D_p	8.192 mm
Heat flux hot channel factor	F_Q	2.60
Effective mass flow rate	$\dot{m}_{eff}$	13457 kg/s
Effective flow area	A_{eff}	3.88 m ²
Coolant inlet temperature	T_{in}	280°C
Rod pitch	r_{pitch}	12.6 mm

Table 1: Westinghouse data

4 Section 1

This section will focus on axial temperature profiles.

4.1 Enthalpy profile

The first step is the axial enthalpy profile that is obtained from the following energy balance:

$$q_{v,tot} A_f dz = W_{HC} di \quad (3)$$

Before integrating this equation, the mass flow rate in the hot channel (W_{HC}) and the total power density received by the coolant ($q_{v,tot}$) have to be computed.

Starting from the Westinghouse data, these steps are performed:

1) Average volumetric heat generation rate:

$$q_{vol,av} = \frac{P_{th} \cdot 0.974}{N_{tot,f,rod} \cdot V_{f,rod}} = \frac{P_{th,core}}{V_{tot,f,rod}} = 355.28 \frac{MW}{m^3} \quad (4)$$

Knowing the average, $q_{v,max}$ can be finally obtained using the heat flux hot channel factor.

$$q_{v,max} = F_Q \cdot q_{vol_{av}} = 923.74 \frac{MW}{m^3} \quad (5)$$

Therefore, the total power density received by the coolant is:

$$q_{v,tot} = \frac{q_{v,HC}}{0.974} \quad (6)$$

2) Average mass velocity:

$$G_{av} = \frac{\dot{m}_{eff}}{A_{eff}} = 3.47 \frac{ton}{m^2s} \quad (7)$$

Since, it is known the geometry of the hot channel, the pitch between fuel rods and the outer-cladding diameter from the Westinghouse data 1, the sub-channel area is computed and then the mass flow rate in the hot channel, with equation 8. Here, it is made the assumption of uniform distribution of the mass flow rate, actually this is not true because of the different power distribution.

$$W_{HC} = G_{av} \cdot A_{sub_{ch}} = 0.304 \frac{kg}{s} \quad (8)$$

Now, all the necessary quantities are known, so, integrating the equation 3, the axial specific enthalpy function in the hot channel is founded.

$$i(z) = i_{in} + 1.0267 \frac{q_{v,max} A_f H_e}{W_{HC} \pi} \left[\sin \left(\frac{\pi z}{H_e} \right) + \sin \left(\frac{\pi H_{act}}{2 H_e} \right) \right] \quad (9)$$

Setting the axial coordinate from $-H_{act}/2$ to $H_{act}/2$.

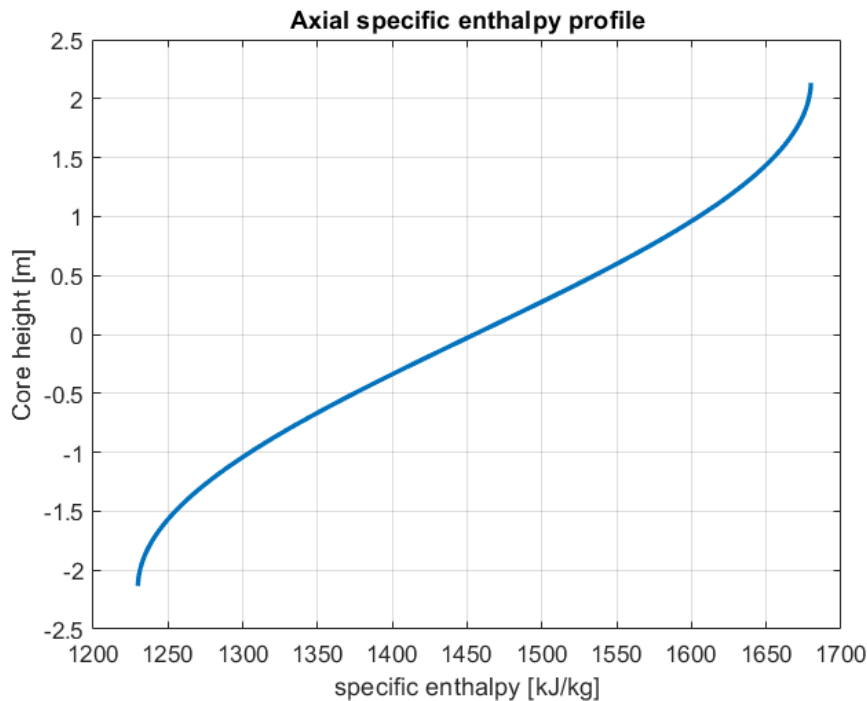


Figure 1: Axial enthalpy profile of coolant in the hot channel

Notice that the saturation enthalpy at 155 bar is reached, then, one expects that at the same point the saturation temperature will also be reached.

4.2 Water profile

Equation 10 is the water temperature function assuming an average specific heat capacity, the coolant profile has the following shape 2. The saturation point is slightly different from the one in the enthalpy profile due to the previous approximation. To have a more precise and consistent coolant profile, the XSteam library is used imposing the pressure and the enthalpy vector. In the end, the height, at which all the water in the hot channel reaches the saturation temperature, is $z_{sat} = 3.36 \text{ m}$ (with 0 at core bottom).

$$T_w(z) = T_{in} + \frac{1.0267 q_{v,max} A_f H_e}{\dot{m}_{HC} \pi \bar{c}_p} \left[\sin\left(\frac{\pi z}{H_e}\right) + \sin\left(\frac{\pi H_{active}}{2H_e}\right) \right] \quad (10)$$

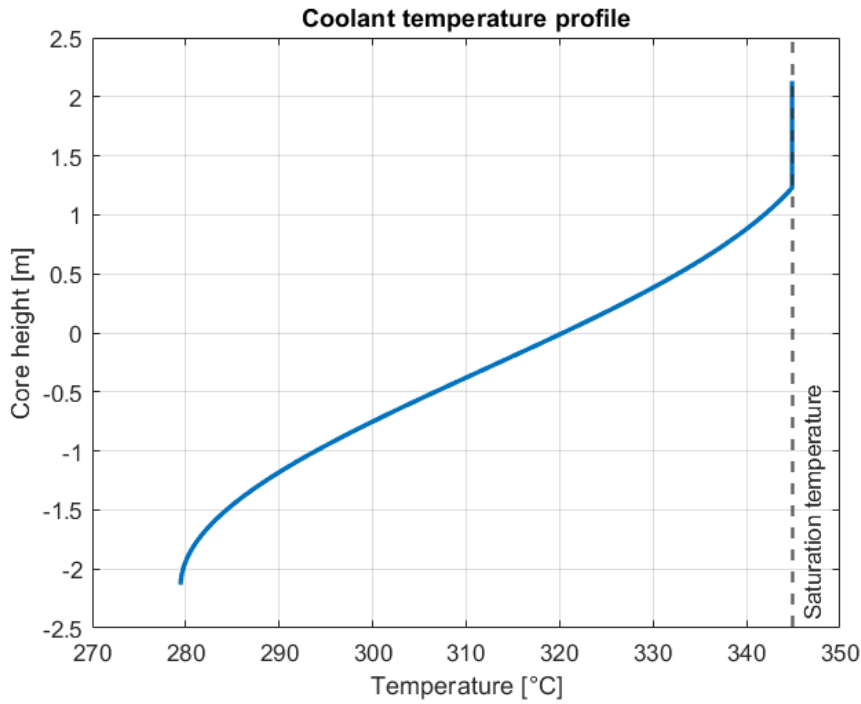


Figure 2: Axial temperature profile of the coolant along the core

4.3 Outer-cladding profile

The heat exchange interaction between the coolant and the outer wall of the cladding is the main focus of this analysis. In fact, the presence of a two-phase flow improves the cooling of the fuel rods but, on the other hand, if uncontrolled, it may be the beginning of DNB accident.

Therefore, the procedure is, using the most suitable correlations, to find the point at which the Onset of Nucleated Boiling (ONB) occurs. The ONB is where the subcooled boiling starts and it talks about subcooled because the boiling is only on the cladding wall, while the bulk water temperature is still below the saturation one. In general, after the onset, there is a region in which the boiling is partially developed; after that, the specific enthalpy continues to increase and the bubbles detach from the wall. until it becomes fully developed. In our case, the saturation temperature is reached, then the

ONB and the detachment will occur for sure. It is made the assumption of considering negligible the region with partial boiling, therefore, from the ONB, it is considered as fully developed boiling. Thanks to this, it is possible to use the Jens-Lottes correlation 11 which, in principle, can describe only the fully developed boiling.

$$T_{co,J-L}(z) = T_{sat} + 25 \cdot q''(z)^{0.25} \cdot \exp\left(-\frac{p_{bar}}{62}\right) \quad (11)$$

Hence, to obtain the total outer-cladding temperature profile, the following steps are performed. Firstly, it is found T_{co} assuming a single-face coolant heat exchange 12. The convective coefficient is retrieved with the Dittus-Boelter equation for turbulent flows 13, assuming the sub-channel closed.

$$T_{co,s-p}(z) = T_w(z) + \frac{q''(z)}{h_l(z)} \quad (12)$$

$$Nu_{D_{eq}} = C (Re_{D_{eq}})^{0.8} Pr^{0.4} \quad (13)$$

Where C takes into account the sub-channel geometry.

Consequently, the ONB is assumed to be where $T_{co,s-p}(z) > T_{co,J-L}(z)$ and from this point the real surface cladding temperature follows the Jens-Lottes function.

The final temperature profile is shown in the figure 3 and it is visible that from the ONB point, the profile is quite flat thanks to the improvement in cooling due to the nucleated boiling.

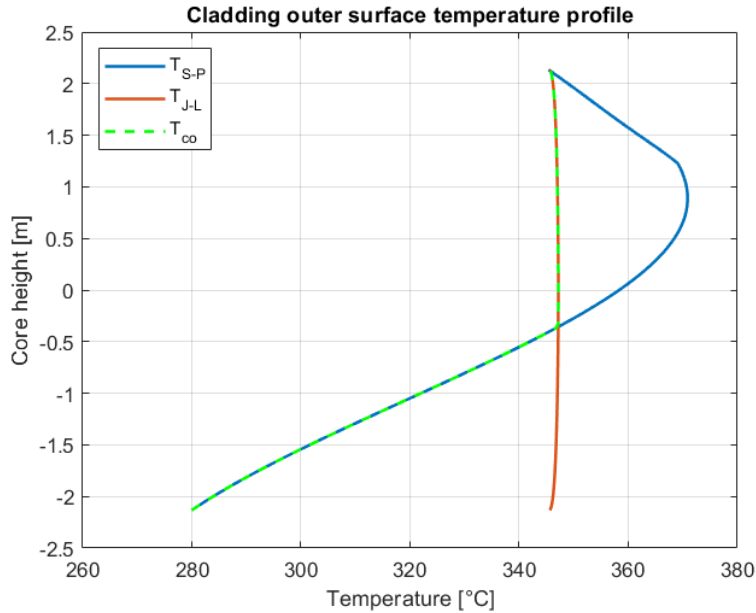


Figure 3: Outer cladding temperature profile, including ONB and boiling effects

4.4 Inner-cladding profile

The evaluation of the inner surface cladding temperature must be performed considering the dependence of the cladding thermal conductivity with respect to the temperature

itself, described in equation 14.

$$k_{cl}(T) = 11.45 + 1.425 \cdot 10^{-2} T \quad (14)$$

Then, from equation 15 the inner temperature of the cladding can be found analytically by solving the integral.

$$\int_{T_{ci}}^{T_{co}} k_{cl}(T) dT = \frac{q_{v,HC}(z) A_f}{2\pi} \ln \left(\frac{D_{co}}{D_{ci}} \right) \quad (15)$$

The figure 4 shows the axial profile of the inner surface of the cladding. It should be noted that there is a change in the concavity of the curve due to the net change in the profile of T_{co} .

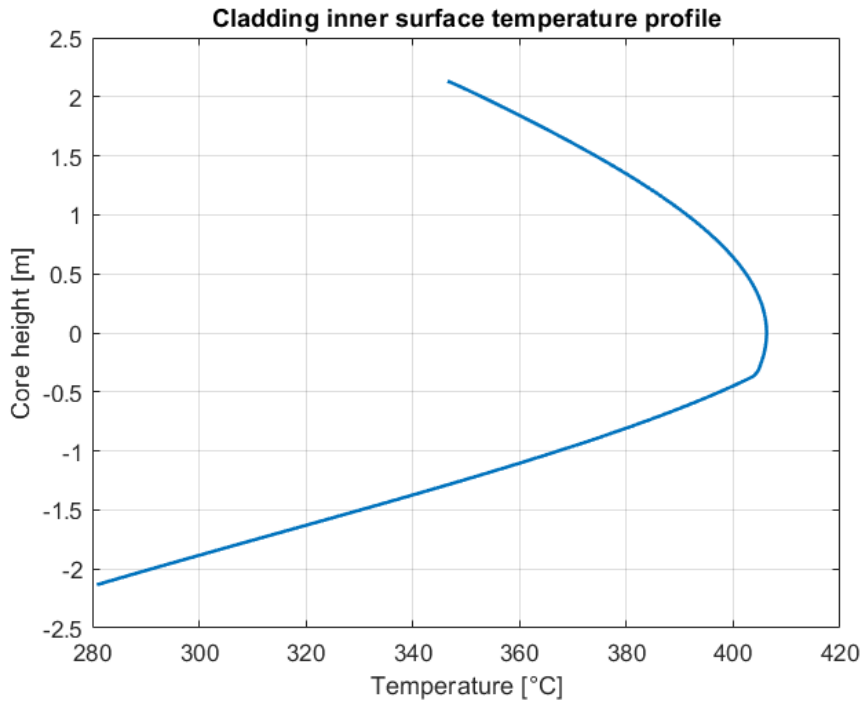


Figure 4: Inner cladding temperature profile

4.5 Fuel profiles

The focus of this section is to calculate the surface temperature of the pellet of UO_2 and, subsequently, the centerline temperature that tells whether it has reached the melting point or not. These steps are implemented in matlab inside a double iterative loop for more reasons.

From Newton's law, the fuel surface temperature is recovered 16.

$$T_{f,S}(z) = T_{ci}(z) + \frac{q_{v,HC}(z) A_f}{\pi D_f h_{gap}} \quad (16)$$

To achieve this scope, one must analyze the behavior of the gas gap between the cladding and the fuel, in particular, its pressure and thermal conductance h_{gap} . It is assumed

to be at BOL, so, there are not gas fission products in the gap that may influence the conductivity but only helium. The gap conductance can be split in three main parts: radiative, convective and conductive.

$$h_{gap} = h_{gas} + h_{rad} + h_c \quad (17)$$

The radiative heat transfer is due to the difference of $T_{f,S}$ and T_{ci} , therefore:

$$h_{rad} = \sigma \frac{T_{f,S}^4 - T_{ci}^4}{T_{f,S} - T_{ci}} \left(\frac{1}{\varepsilon_f} + \frac{1}{\varepsilon_{cl}} - 1 \right)^{-1} \quad (18)$$

The convective part takes into account the convection with the surfaces and is the main component of the total gap conductance. It is described with Ross and Stoute correlation 19. Where R are the material's roughness and l the extrapolated length

$$h_{gas} = \frac{k_{gas}}{\beta^2 R_f + R_{cl} + \delta + l_f + l_{cl}} \quad (19)$$

The δ is the gap thickness and is strictly related to the last component of the gap conductance. This parameter is influenced by the elastic deformation of the cladding due to the pressure gradient and by the thermal expansion of both pellet and cladding.

Starting from the cladding, the two phenomena are treated separately one from the other. The diameter of the cladding is reduced because the outer pressure (nominal system pressure) is 155 bar and the internal pressure reaches a maximum value of 84 bar, so it is compressed 20. On the other hand, the thermal expansion tends to work in the opposite direction; the overall result is a slight increase in the cladding diameter of the order of micrometer.

$$\frac{\Delta r_{ci}}{r_{ci}} = \frac{1}{E(T)} \left[\frac{1}{\gamma^2 - 1} p_{in}(1 - \nu) + (1 + \nu) \frac{\gamma^2 - 2}{\gamma^2} p_{bar} \right] \quad (20)$$

The pellet expansion is computed like the cladding one 21, but its variation is much larger, leading to a contact between the pellet and the cladding. When this happens, the farther contribution in the gap conductance is obtained with the Dean correlation 22, knowing the surface contact pressure p_c and the constant $b = 1.36 \cdot 10^{-3} \frac{W}{m^2 C}$.

$$D_p^T - D_p^{T_a} = D_p^{T_a} \alpha_f (\bar{T}_f - T) \quad (21)$$

$$h_c = b p_c \quad (22)$$

Here, the first reason for an iterative process, in equation 21, the pellet diameter depends on the average fuel temperature in the radial direction, therefore to calculate $T_{f,S}$ also the fuel radial profile is needed. The heat conduction equation in cylindrical coordinates, neglecting the axial conduction, is the next one 23.

$$\frac{d^2 T}{dr^2} + \frac{1}{r} \frac{dT}{dr} + \frac{q_{v,HC}}{k_{UO_2}(T)} = 0 \quad (23)$$

It is an ordinary differential equation of the second order but non linear due to the dependance of the uranium oxide thermal conductivity from the temperature.

Hence, the iterative process is made as follow:

1. Guess of h_{gap} to initialize a value for $T_{f,S}$.
2. Analytical resolution of the equation 23, considering a guessed average value for the thermal conductivity, obtaining:

$$T_f(r) = T_{f,S} + \frac{q_{v,HC} \cdot R}{4 \bar{k}_{UO_2}} \left(\frac{D_p^2}{4} - r^2 \right) \quad (24)$$

3. Using the Westinghouse correlation for the thermal conductivity 25, it is made an integral average with the matlab function *trapz*.

$$k_{UO_2}(T) = \frac{1}{11.8 + 0.0238T + 8.775 \cdot 10^{-13}T^3} \quad (25)$$

4. With the new value of $\bar{k}_{UO_2}$, the radial fuel profile is calculated again, finding the fuel centerline temperature with $T_{f,C} = T_f(r = 0)$.
5. Having the radial profile, also in this case, $\bar{T}_f$ is done as the integral average weighted with the area.

$$\bar{T}_f = \frac{\int_0^{R_p} T_f(r) \cdot 2\pi r \, dr}{\pi R_p^2} \quad (26)$$

6. Inside the inner loop, with $\bar{T}_f$ and the mean temperature gap, all the step to compute h_{gap} are performed with the check if or not there is the pellet-cladding contact.
7. Check the relative error of $T_{f,S}$ with respect to the previous iteration and the same in the outer loop for $T_{f,C}$, with a minimum tolerance of 10^{-6} .

The results with the axial profiles are showed in figure 5 and 6. The surface temperature has an hourglass shape which is typical of contact between cladding and fuel. In fact, in the middle of the core there is the higher power generation, therefore the higher temperature; leading to a much larger pellet thermal expansion that touches the cladding locally. Where there is contact, gap thickness goes to zero, therefore the gap conductance increases thanks to the additional conductive component and also because the main component h_{gas} increases, improving the heat exchange between pellet and cladding, thereby a local temperature depression.

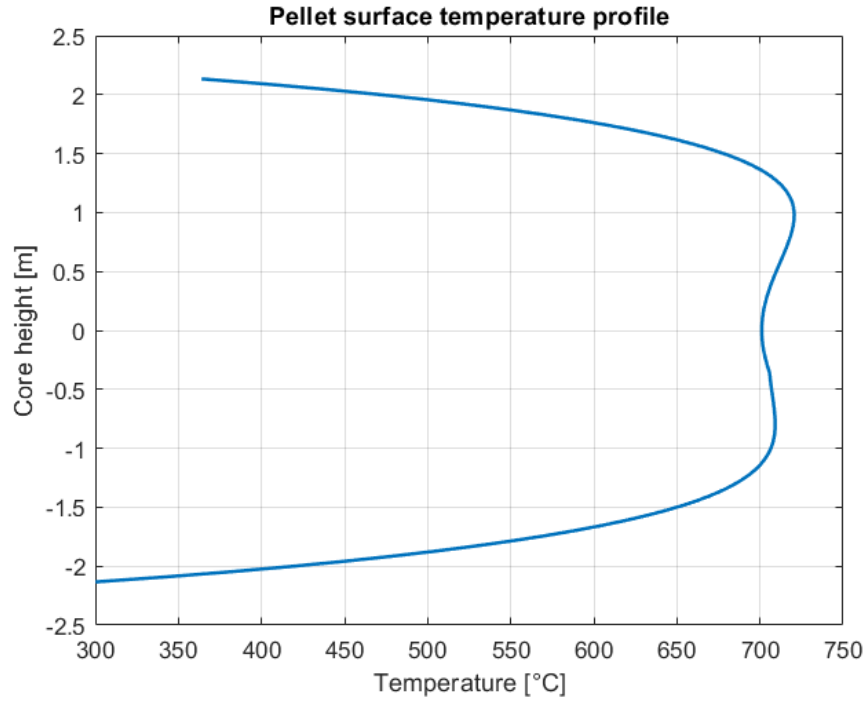


Figure 5: Axial profile of fuel surface temperature with contact effects

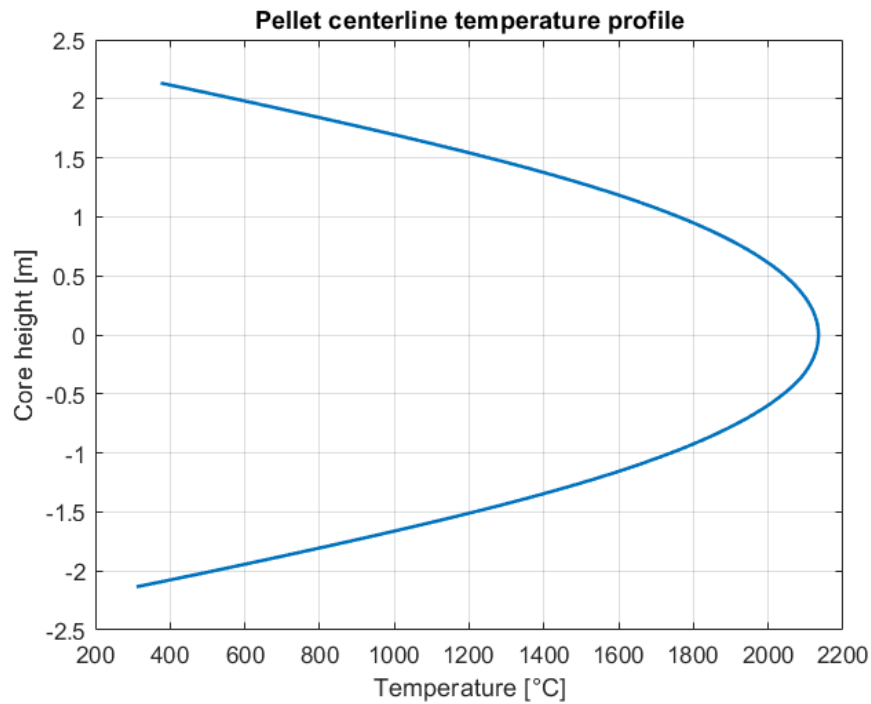


Figure 6: Axial profile of fuel centerline temperature with contact effects

The centerline temperature has a shape consistent with the chopped cosine distribution of the power density generated inside the pellet, with the peak value in the middle. Additionally, the surface temperature reduction has a direct consequence also in the peak centerline temperature, equal to $2134\text{ }^{\circ}\text{C}$, which is lower with respect to the possible value

without expansion, around of $2450\text{ }^{\circ}\text{C}$. The peak value is what must be compared with the melting temperature of the sintered pellet of uranium dioxide with a porosity of 4.5%, that is assumed $2800\text{ }^{\circ}\text{C}$, therefore the thermal limit is respected.

One must be aware that it was made the assumption of normal operation in steady-state. There may be same transient in which the power is higher or the cooling is worst and then the peak centerline temperature could reach higher values. To ensure the maximum safety standard, the thermal limits must be analyzed in the worst case scenario.

The figure 7 collects all the computed axial temperature profiles. An evident feature is the large temperature gradient inside the pellet, mostly cause by the lower thermal conductivity that is about $1/4$ of the cladding one. The last three profiles follow a chopped cosine shape, since the analysis started with that distribution of the power density. Although the temperature at the top of the core is larger in all cases because the coolant temperature is hotter at the outlet, then the smaller temperature gradient between the water and the outer surface of the cladding, reduces the cooling capability.

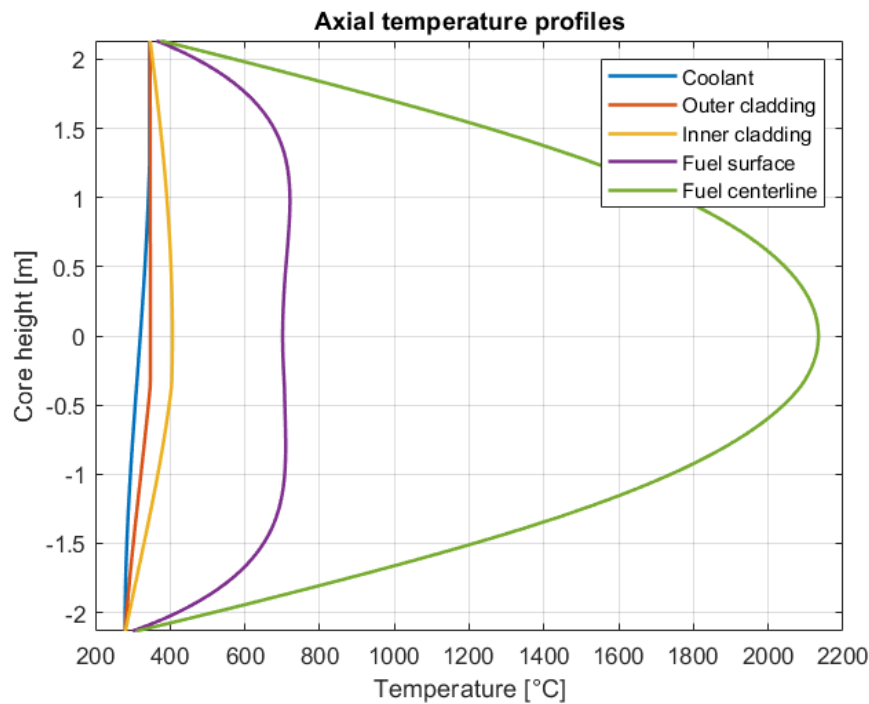


Figure 7: Comparison of all axial temperature profiles

In figure 8 the temperature difference between water and cladding is more evident. It should be remember that in a PWR, boiling water must be strongly avoided and, in our case, the saturation temperature was reached, so that is a critical region for the boiling crisis. All the heat flux coming from the fuel is used for the phase change of the water, therefore is mandatory to find the Critical Heat Flux and the MDNBR, understanding how far is the DNB. This is made in the next section.

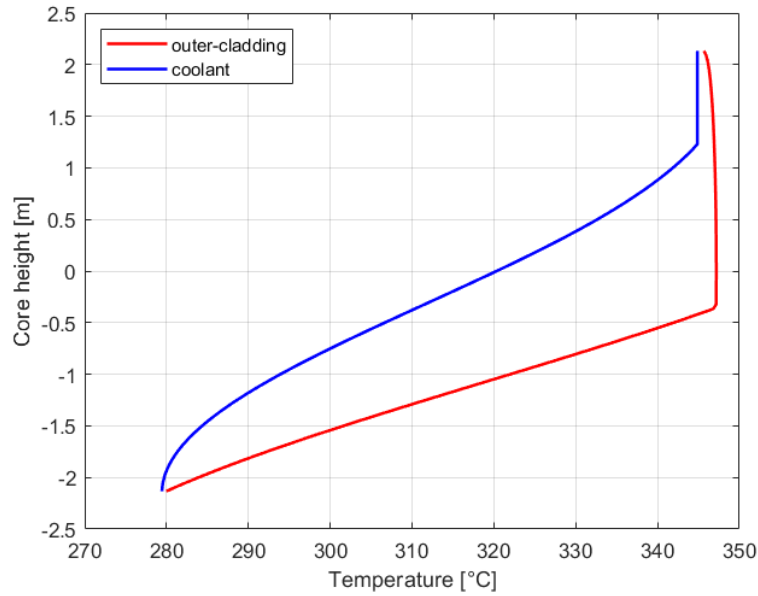


Figure 8: Coolant-to-cladding temperature difference

5 Section 2

The CHF is the flux that causes a total evaporation of the first liquid film that cooled the cladding, not allowing other liquid part to reach the wall. The heat conductance of the vapor is much lower than the liquid one, leading to an higher thermal resistance; then the heat starts to accumulate inside the rods increasing the temperature with consequently larger thermal stresses and possible failure of the first two barriers of the reactor. Therefore, the DNBR, defined below 13, is one of the parameters that quantify the safety of a nuclear plant.

$$DNBR = \frac{q_{c,NU}}{q} \quad (27)$$

The task is quite challenging because the two-phase flow must be studied up to the precision of the single bubble attached to the cladding walls and in reality are used a bunch of semi-empirical correlations which could be different according to working conditions, geometry, simplifying assumptions.

The main steps are the computation of the flow quality, tightly related to the void fraction and the critical heat flux considering the non uniformity of the flux itself.

5.1 Equilibrium flow quality

A first good evaluation of the flow quality, knowing the specific enthalpy profile, is that one imposing the thermodynamic equilibrium. From XSteam, the saturation values of the enthalpy are extracted and from the equation 28 is obtained the equilibrium quality profile shows in figure 9.

$$x_{eq}(z) = \frac{i(z) - i_{ls}}{i_{vs} - i_{ls}} \quad (28)$$

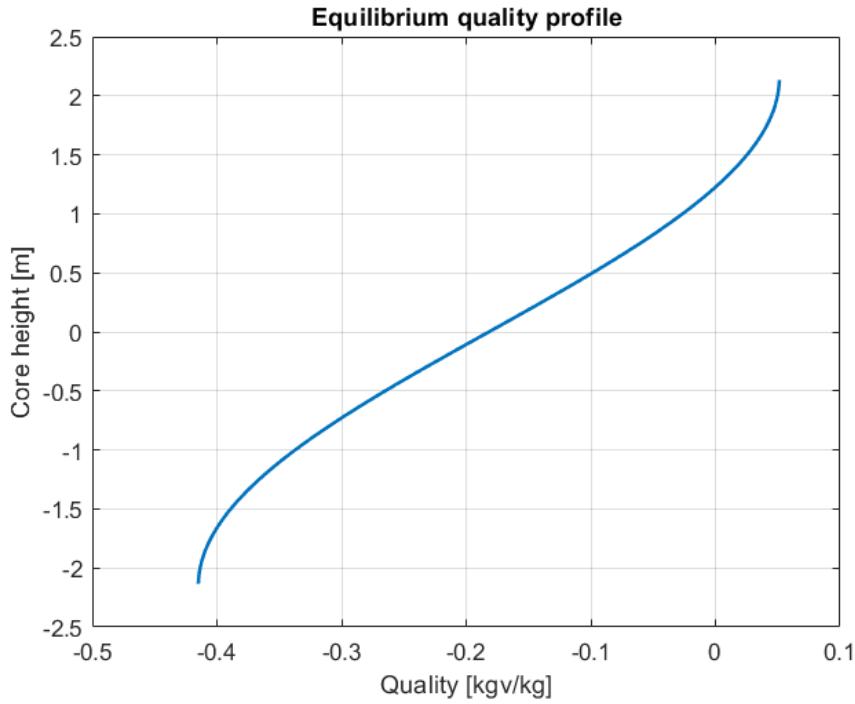


Figure 9: Equilibrium quality profile

As expected, the quality is mostly negative, since, up to 3.36 meter, the enthalpy of the water is below the saturated liquid. After that point, the quality reaches a maximum value of 5.1%.

By definition, this way to compute the quality assumes a uniform enthalpy in each axial point, but there is the possibility of subcooled boiling with the creation of bubble where there is a locally increase of the enthalpy, that is the surface of the cladding. Hence, the formula 29 is more precise as to what the real steam quantity is inside the sub-channel. Of course, before detachment, there is no steam in the main flow, therefore x is equal to zero.

$$x = \frac{\dot{m}_v}{\dot{m}_v + \dot{m}_l} \quad (29)$$

To retrieve this quantity, the Bowring model, based on experiments, is used. In this correlation, the heat flux received from the cladding is subdivided as follows 30:

- Latent heat of evaporation q_e
- Convection caused by bubble agitation q_a
- Single-phase heat transfer between patches of bubbles q_{sp}
- Recondensation of the bubbles at the wall q_{cond}

$$q = q_e + q_a + q_{sp} + q_{con} \quad (30)$$

The recondensation is assumed negligible and to be consistent, since in the previous section the partial developed boiling was considered negligible, here, q_{sp} is imposed equal to zero. Then from the power balance, the resolution of the integral 31 allows to find the most realistic flow quality.

$$x = \int_{z_d}^z \frac{p_H q''}{i_{evap} W_{HC} (1 + \varepsilon)} dz \quad (31)$$

The missing parts are the coefficient $\varepsilon = \frac{q_a}{q_e}$ and the detachment point. The coefficient $1 - \varepsilon$ tells how much of the flux is useful to evaporate the water and can be found with the Rouhani correlation 32. The evaporation part, after the detachment, is always above 60%.

$$\varepsilon = \frac{\rho_l}{\rho_{g,sat} i_{evap}} (i_{l,sat} - i_l) \quad (32)$$

The detachment point is found with the Griffith model 33, when $T_w > T_{l,D}$ and in our case is equal to $z_D = 2.93 \text{ m}$.

$$T_{l,D}(z) = T_{sat} - \frac{q''(z)}{5 h_l(z)} \quad (33)$$

The figure 10 compares the quality profile from detachment to outlet with the two different approaches. It is evident what it has been said, that is, when the equilibrium enthalpy is below the saturated liquid ($x_{eq} < 0$), subcooled boiling is possible anyway. This is what actually happens, with the B-R quality profile shifted on the right and with a maximum value of 7.5% of steam.

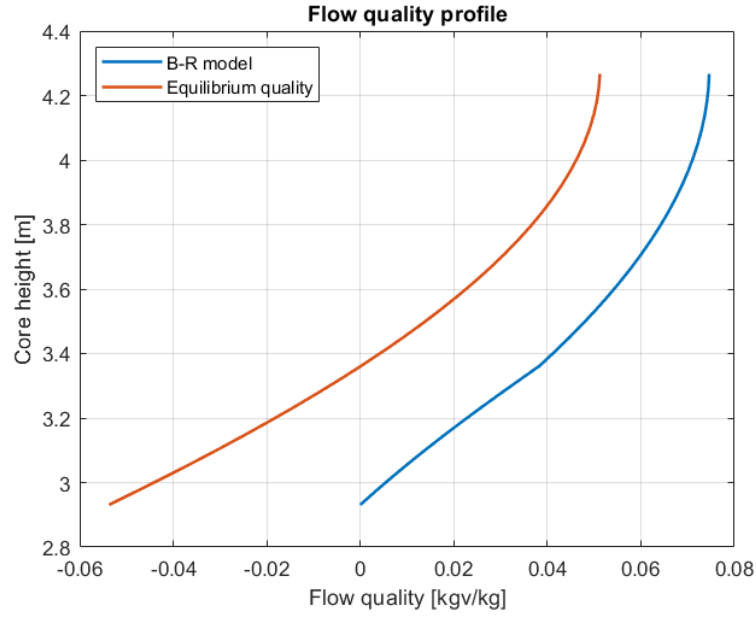


Figure 10: Comparison between equilibrium and Bowring-Rouhani quality profiles

For completeness is also derived the void fraction profile 11 with a null value up to the ONB, from this point to detachment the rate of bubbles growth on the wall is assumed linear and from detachment is evaluated by the Zuber-Findlay correlation 34. Note that although the maximum quality is 7.5%, the maximum void fraction is about 28% of the whole volume of the subchannel.

$$\alpha = \frac{\frac{x}{\rho_g}}{C_0 \left(\frac{x}{\rho_g} + \frac{1-x}{\rho_l} \right) + C_1 \cdot \frac{1}{G} \left(\frac{\sigma g (\rho_l - \rho_g)}{\rho_l^2} \right)^{0.25}} \quad (34)$$

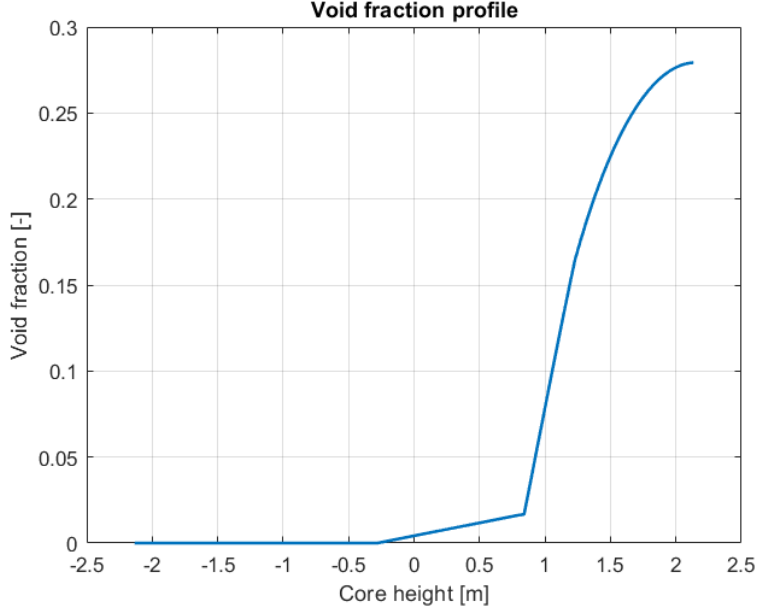


Figure 11: Void fraction profile along the core height

5.2 Critical Heat Flux

For the evaluation of the critical heat flux, Westinhouse developed a series of correlations. Here, the W-3 correlation by Tong is used, but it is built for a 15x15 fuel assembly and assuming uniform heat flux; therefore, it is applied a correction factor of 0.88 since the AP1000 assemblies are 17x17 and to take into account the cosine shape of the flux is used the Tong F-factor. Before that, the uniform critical heat flux for 17x17 assemblies $q_{c,EU}$ multiplies the grid spacer factor F_S given that the spacer grid promotes turbulence, increasing heat exchange and decreasing the CHF.

The Tong factor F considers the non-uniformity of the flux considering the propagation of heat from a zone with higher flux that goes toward the outlet with an exponential decreasing of importance. Below, the equations 35 and 36 are the formulae used. In this case, the equilibrium flow quality is used because it is the actual quantity in the correlation W-3.

$$F = \frac{\int_0^{L_{c,NU}} q(z') \exp[-C(L_{c,NU} - z')] dz'}{q_{local} (1 - \exp(-C \cdot L_{c,NU}))} \quad (35)$$

$$C = 0.15 \cdot \frac{(1 - x_{eq}) L_{c,NU}}{4.31 G^{0.478}} \quad (36)$$

The Tong factor avoids an overestimation of non-uniform critical heat flux ($F > 1$), in fact, as seen in equation 37, is at the denominator.

$$q_{c,NU} = \frac{q_{c,EU}}{F} \quad (37)$$

The figure 12 shows all the three CHF's computed, compared with the heat flux deposited in the water. The important feature is the Tong factor effect, in fact, with respect to the red dashed line, it can be seen that the CHF non-uniform decreases in the upper core region even though the heat flux is lower, precisely because the water is heated more in the center where there is an higher heat flux and by advection the coolant keeps memory of this. Another aspect is that the CHF is always above the real heat flux but to assert that the thermal limit of MDNBR is respected, the ratio between these two quantities must be done 27.

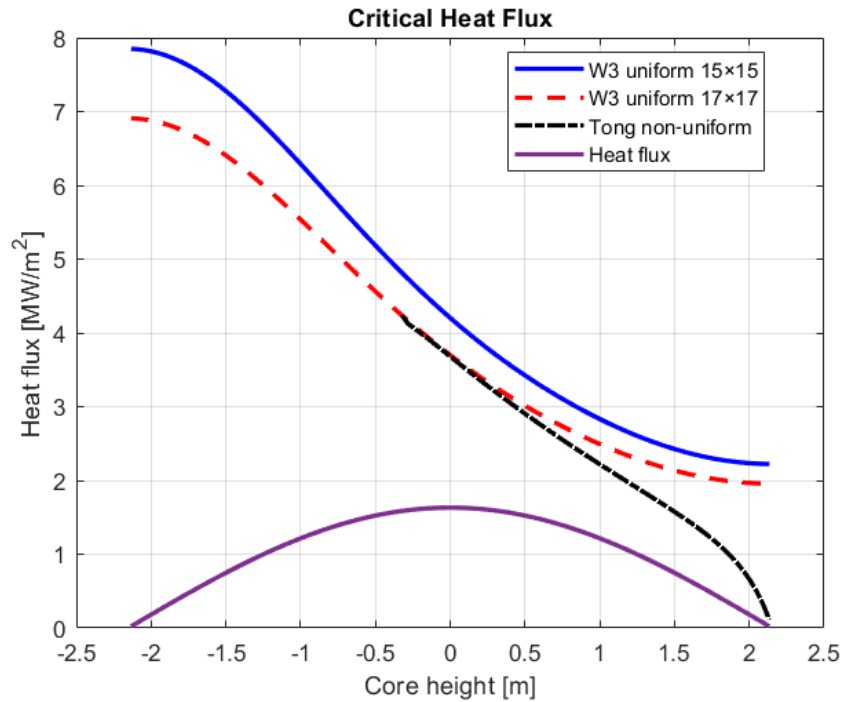


Figure 12: Critical Heat Flux profiles compared with the actual heat flux

Finally, the last figure 13 represents the DNBR profile starting from the ONB point. The first characteristic to notice is that the minimum DNBR found 1.82 is lower than the threshold setting as the safety margin of 1.85. Therefore, the thermal limit is not respected; however, many assumptions were made, so it is something expected. It is interest to analyze where the MDNBR occurs, that is at $z = 3.02 \text{ m}$, just after the detachment point in the subcooled boiling region, the worst zone in which the DNB can happens. Looking at the figure 12, one may think that the MDNBR should be at the top core because the CHF is lower, actually, in that zone the heat flux is two order of magnitude lower with respect to the center, in fact, the DNBR reaches the peak value.

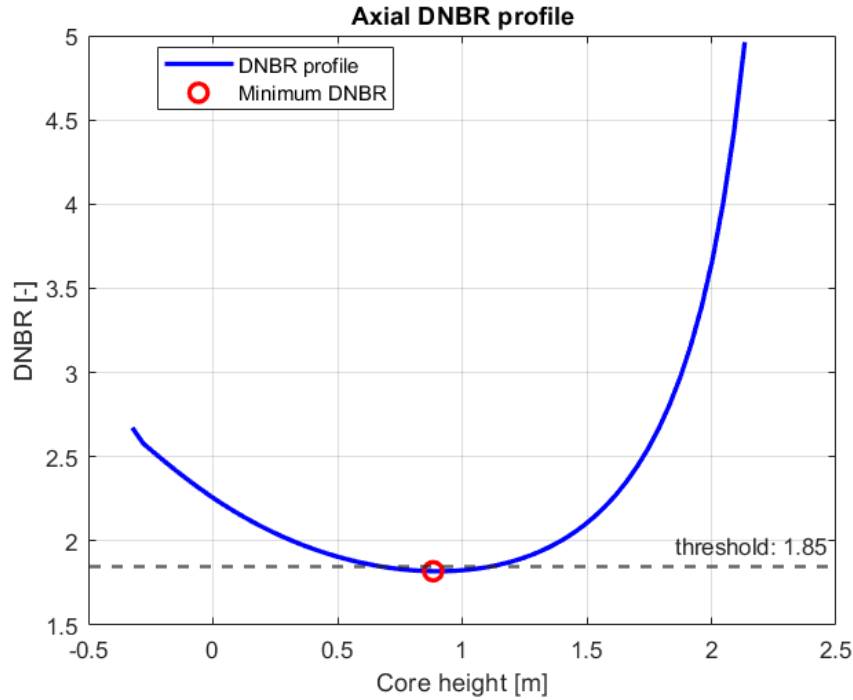


Figure 13: Axial DNBR profile with minimum DNBR and safety threshold

6 Conclusion

The thermal-hydraulic analysis confirms that the fuel centerline temperature remains below the safety limit, with a peak value of $2134\text{ }^{\circ}\text{C}$ compared to the $2800\text{ }^{\circ}\text{C}$ melting point. However, the minimum DNBR found is 1.82, slightly below the threshold of 1.85, indicating a potential risk of DNB around 3.02 m , in the subcooled boiling region.

It is important to note that the W-3 correlation is based on equilibrium quality x_e , which may overestimate the CHF as it does not account for local bubble dynamics. In fact, using the Bowring-Rouhani correlation, the actual quality reaches 7.5%, suggesting a less conservative result.

Moreover, the mechanical contact between pellet and cladding in the mid-core region improves heat transfer and reduces surface temperature, but introduces localized stress. Also the assumption of uniform flow distribution and the simplified boiling model likely contribute to the DNBR value and temperature profiles.